

Naive Bayes Algorithm



What is Naive Bayes?

Naive Bayes is a **supervised machine learning algorithm** mainly used for:

- Classification problems
- Spam detection
- Sentiment analysis
- Text classification
- Medical diagnosis

It is based on **Bayes' Theorem** and assumes that all features are independent.

Bayes Theorem

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$$

Where:

Symbol	Meaning
$P(A B)$	Probability of A given B
$P(B A)$	Probability of B given A
$P(A)$	Probability of A
$P(B)$	Probability of B

Why is it Called “Naive” ?

The algorithm assumes:

Every feature is independent of every other feature.

Example:

If predicting whether a fruit is an apple using:

- Color
- Size
- Taste

Naive Bayes assumes these features are unrelated.

Types of Naive Bayes

1. Gaussian Naive Bayes

Used for continuous numerical data.

Examples:

- Height
 - Weight
 - Temperature
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2. Multinomial Naive Bayes

Used for discrete counts.

Examples:

- Word frequency
 - Text classification
-

3. Bernoulli Naive Bayes

Used for binary data.

Examples:

- Yes/No
- True/False
- 0/1

Working of Naive Bayes

Steps

1. Calculate prior probabilities
2. Calculate likelihood probabilities
3. Apply Bayes theorem
4. Choose the class with highest probability

Example — Spam Email Detection



Suppose we have the following dataset:

Email	Offer	Win	Spam
1	Yes	Yes	Spam
2	Yes	No	Spam
3	No	Yes	Spam
4	No	No	Not Spam
5	No	No	Not Spam

Now a new email arrives:

"Win Offer"

We want to classify it.

Step 1 — Prior Probability

Total emails = 5

Spam emails = 3

Not Spam emails = 2

$$P(\text{Spam}) = \frac{3}{5}$$

$$P(\text{NotSpam}) = \frac{2}{5}$$

Step 2 — Likelihood Probability

For Spam:

Feature	Probability
Offer	2/3
Win	2/3

For Not Spam:

Feature	Probability
Offer	0/2
Win	0/2

Step 3 — Apply Bayes Theorem

For Spam:

$$P(\text{Spam}|\text{Offer}, \text{Win})$$

$$= P(\text{Spam}) \times P(\text{Offer}|\text{Spam}) \times P(\text{Win}|\text{Spam})$$

$$= \frac{3}{5} \times \frac{2}{3} \times \frac{2}{3}$$

$$= 0.266$$

For Not Spam:

$$= \frac{2}{5} \times 0 \times 0$$

$$= 0$$

Final Prediction

Since:

$$0.266 > 0$$

The email is classified as:

 **Spam**

Fruit Classification Example



Dataset:

Fruit	Color	Shape
Apple	Red	Round
Banana	Yellow	Long
Apple	Red	Round
Banana	Yellow	Long

New Fruit:

- Red

- Round

Prediction:

✓ Apple

Advantages of Naive Bayes

- Fast and efficient
 - Simple to implement
 - Works well with large datasets
 - Good for text classification
 - Requires less training data
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Disadvantages

- Assumes feature independence
 - Zero probability problem
 - Less accurate for complex relationships
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Laplace Smoothing

Used to avoid zero probability.

Formula:

$$P = \frac{Count + 1}{Total + Number\ of\ Classes}$$

Applications of Naive Bayes

- Email Spam Filtering
- Sentiment Analysis
- Medical Diagnosis

- Recommendation Systems
- News Classification

Comparison with Other Algorithms

Algorithm	Speed	Accuracy	Best For
Naive Bayes	Very Fast	Good	Text data
Decision Tree	Medium	Good	Rule-based data
SVM	Slow	High	Complex classification
KNN	Slow	Medium	Small datasets

Python Example


```
from sklearn.naive_bayes import GaussianNB
import numpy as np

# Features
X = np.array([
    [1, 1],
    [1, 0],
    [0, 1],
    [0, 0]
])

# Labels
y = np.array([1, 1, 1, 0])

# Create model
model = GaussianNB()

# Train model
model.fit(X, y)

# Predict
prediction = model.predict([[1, 1]])

print(prediction)
```